



The Challenge in Engineering Higher Education

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- ❖ About Afeka
- ❖ Personal perspective
- ❖ The central problem
- ❖ The drivers of change
- ❖ Future challenges
- ❖ Team model
- ❖ The profile of the future "lecturer"

Afeka Academic College of Engineering in Tel Aviv

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- 8-B.Sc Programs
- 5-M.Sc Programs
- Graduate Profile
- The Educational Mode
- The Afeka Learning Experience



4



An algorithm with self-confidence
AI - The perfect lecturer

- **Very confident – even when wrong**
 - **It's not responsible when things break in production**
 - **No human interaction**
- Available 24/7
 - Never says “this is trivial”
 - Explains again without getting annoyed
 - Very confident – even when wrong
 - It computes faster than us
 - It writes code without complaining
 - It never asks why we need this

Drivers of change

5



Apps Library Genesis Directory of all Conf... Conference Alerts 2... Exploration of Teach... Education, Research... ICOF2016 Imported From Edge ChatGPT

הנדסת חשמל א Shared

Sources

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Try Deep Research for an in-depth report and new sources!

Search the web for new sources

Web Fast Research

Select all sources

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Chat

7 8

- **Electric Voltage (V)**: This represents the potential difference or work required to move a charge between two points ($V = \frac{dW}{dq}$), measured in **Volts (V)** 9.
- **Resistance (R) and Conductance (G)**: Resistance is a material's opposition to current, governed by its resistivity (ρ), length (l), and cross-sectional area (A): $R = \rho \frac{l}{A}$ 10 11. Conductance is the inverse of resistance ($G = \frac{1}{R}$) 11.

3. Circuit Components and Analysis Context

In the larger context of **circuit analysis**, these concepts are applied to various components that either consume, store, or provide energy:

- **Passive Components**:
 - **Resistors (R)**: Dissipate energy as heat according to **Ohm's Law** ($v(t) = R \cdot i(t)$) 11 12.
 - **Capacitors (C)**: Store energy in an electric field 13. The relationship between charge and voltage is $q = C \cdot V$ 14.
 - **Inductors (L)**: Store energy in a magnetic field created by current flow 15 16.
- **Governing Laws**: Analysis is predicated on **Kirchhoff's Laws**:
 - **KCL (Current Law)**: The sum of currents at any node equals zero 17.
 - **KVL (Voltage Law)**: The sum of voltages in a closed loop equals zero 18.

4. Engineering Classification: Lumped vs. Distributed Systems

For the purpose of analysis, electrical engineers classify circuits based on the relationship between the physical size of the components and the signal's wavelength:

- **Lumped-Element (ורכיבים מקובצים)**: When the physical dimensions are negligible compared to the wavelength, allowing for simplified analysis using algebraic and ordinary differential equations 19 20.
- **Distributed-Element (ורכיבים מפוזרים)**: When the physical size is comparable to the wavelength, requiring more complex analysis as current and voltage vary along the length of the conductors 21.

Start typing... 10 sources

Studio > Video Overview

Demystifying Circuits

Based on 10 sources

Demystifying Circuits

00:00 08:52

1x

Good video Bad video

Demystifying Circuits

- Teamwork
- Flipped classroom
- Personalized learning
- Students teach
- AI (tool to improve learning processes and help with problem solving)

ADSP-Course

Project/Problem
Based Learning
(PBL)

Principles of
designing real-
time algorithms

How do you plan and manage the course/lesson?



How can learning processes be assessed, evaluated, and continuously improved?

Fostering Motivation for Long-Term, Meaningful Learning.

How do we fairly assess individual contributions in team-based projects?

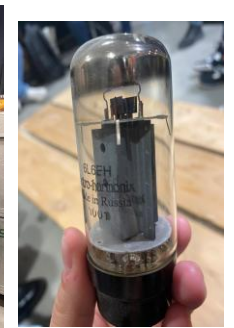
I may fail to achieve the goals?????



Learning/Teaching Process

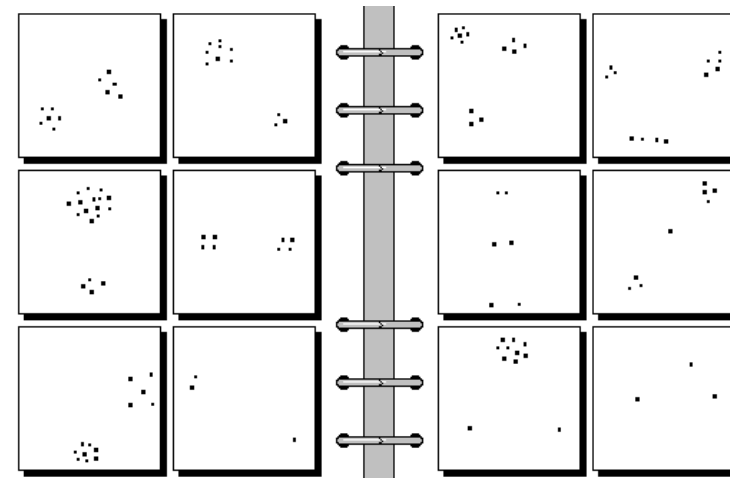
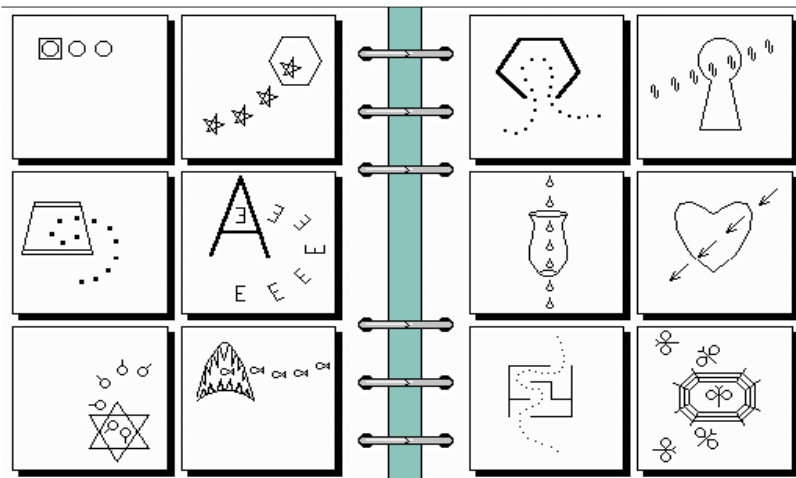
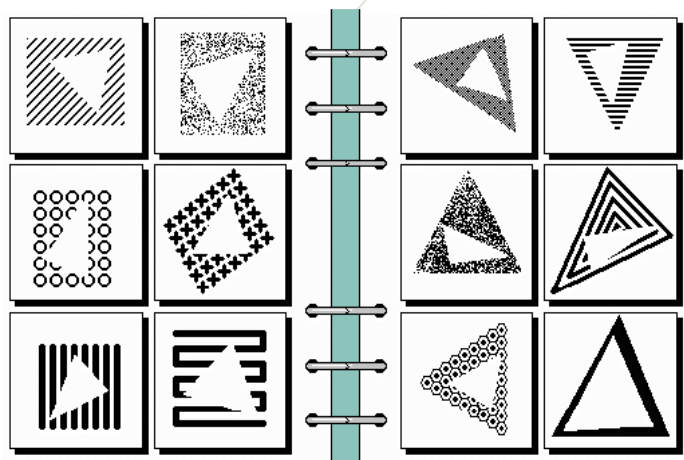
7

- Adaptive and Dynamic Course Design
- Instructor Availability
- Pre-assessment Questionnaire
- Each lecture begins with a problem to solve
- Students are required to prepare a presentation
- Pre-class preparation questionnaire for the next session (presentation, video, instructional content)
- Assessment of learning outcomes
- Discussion
- Bridging knowledge gaps
- Immediate formative feedback (Delta–Plus)
- Industry site visits and guest experts from industry



Mikhail Moiseevitch Bongard

Проблема Узнавания (The Problem of Recognition) 1967



Examples of student work in the ADSP course (DTFT)

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Discrete-Time Fourier Transform Lecture Creation

Sources

+ Add sources

Try Deep Research for an in-depth report and new sources!

Search the web for new sources

Chat

The sources contain a specific directive regarding the c

Under the heading "DTFT", the instruction given is to C

Discrete-Time Fourier Transform 1 .

Save to note

Demystifying the DTFT



Start typing...

1 source

Unlocking Digital Signals



Instruction or directive 1 .

ate lecture about DTFT 1 . Based on the
which stands for Discrete-Time Fourier

What is the lecture topic?

00:59

06:02

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Good video

Bad video

NotebookLM can be inaccurate; please double check its responses.

Insights from personal experience

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- Post-hoc grading alone has little educational impact
- Can AI teach? It does not work effectively in large groups
- Lecturers are irreplaceable — yet teaching and learning practices must evolve
- Policy message: education requires more investment, not less
- Lecturers must invest substantial effort
- Recognizing the strengths and limitations of AI tools is critical in course design
- Design learning challenges beyond the capabilities of AI tools

The Structural Challenge Higher Education

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- ☐ Increasing workload and role overload for engineering faculty
- ☐ A uniform lecturer role misaligned with a changing engineering landscape
- ☐ Negative impact on teaching quality, innovation, and faculty sustainability
- ☐ Structural change is required — not incremental adjustment

The Proposed Principle

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- ❑ Shift from a single-lecturer model to a complementary team-based model
- ❑ Unbundling and realignment of academic roles
- ❑ Targeted integration of teaching, PBL, industry engagement, AI, and faculty development

Uniformity does not create excellence. Complementary strengths do.

Regulatory Implications

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- ☐ Formal openness to recruiting industry professionals and hybrid profiles
- ☐ Differentiated academic career pathways
- ☐ System-level quality indicators: learning experience, graduate readiness, faculty sustainability

The future is not the perfect lecturer—it is a well-designed academic ecosystem.

“Logic takes care of itself; all we have to do is to look and see how it does it”. Ludwig Wittgenstein

Thank You